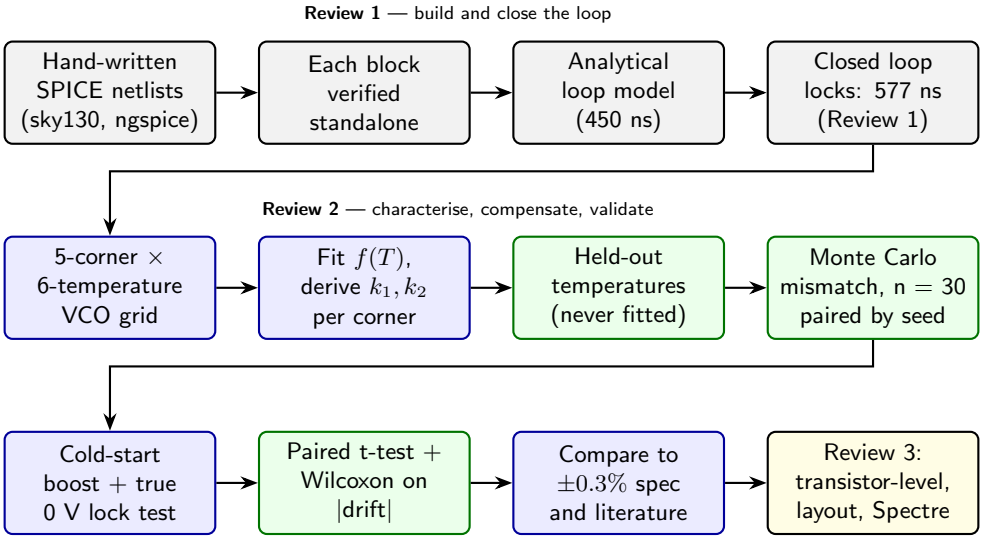

Review 1 — build and close the loop
Hand-written SPICE netlists (sky130, ngspice)
Each block verified standalone
Analytical loop model (450 ns)
Closed loop locks: 577 ns (Review 1)
Review 2 — characterise, compensate, validate
5-corner × 6-temperature VCO grid
Fit $f(T)$, derive k_1, k_2 per corner
Held-out temperatures (never fitted)
Monte Carlo mismatch, n = 30 paired by seed
Cold-start boost + true 0 V lock test
Paired t-test + Wilcoxon on |drift|
Compare to $\pm 0.3\%$ spec and literature
Review 3: transistor-level, layout, Spectre
grey = Review 1 blue = design step green = validation step (guards against fooling ourselves)